MIL-S-18568B(OS)
7 August 1972
SUPERSEDING
MIL-S-18568(NORD)
5 May 1955

MILITARY SPECIFICATION

SIGNAL, ILLUMINATION, MARINE, MK 1 MODS 0, 1, 2, 3, and 4 AND SIGNAL, SMOKE, MARINE, MK 2 MODS 0 AND 1

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope. This specification covers the requirements and preparation for delivery of the Signal, Illumination, Marine, Mk 1 Mods 0, 1, 2, 3, and 4 and Signal, Smoke, Marine, Mk 2 Mods 0 and 1 and the methods of inspection and tests upon which product acceptance shall be based. The signal consists of an aluminum cartridge case containing the rocket assembly and pyrotechnic assembly. The signal is classified as class B explosives (Code of Federal Regulations 49 CFR 171-178) and class II C explosives (Coast Guard Rules and Regulations for Military Hazardous Munitions CG 108).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-45662	Calibration System Requirements

FSC 1370

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:43Z
Check the source to verify that this is the current version before use.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection By Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for

DRAWINGS

Bureau of Naval Weapons

LD 615236	Pack, Shipping and Storage, Signal, Illumination, Marine, Assembly
LD 615246	Signal, Illumination, Marine, Mk 1 Mod 1 (Star)
LD 615259	Package, Signal, Illumination, Marine
LD 615260	Signal, Illumination, Marine, Mk 1 Mod 0 (Comet)
LD 615261	Signal, Illumination, Marine, Mk 1 Mod 2 (Chameleon)
LD 615262	Signal, Illumination, Marine, Mk 1 Mod 3 (Shower)
LD 615263	Signal, Illumination, Marine, Mk 1 Mod 4 (Occulting, Chameleon)
LD 615264	Signal, Smoke, Marine, Mk 2 Mod 0
LD 615265	Signal, Smoke, Marine, Mk 2 Mod 1

(And all LD's, drawings, specifications, standards, and publications listed thereon.)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:43Z
Check the source to verify that this is the current version before use.

should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Code of Federal Regulations

49 CFR 171-178	Hazardous Materials Regulations, Department of Transportation

Coast Guard Rules and Regulations

CG 108	Rules and Regulations for Military Explosives and Hazardous Munitions

(Applications for copies of the above publications should be addressed to Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Inspection and manufacture. The signal shall be manufactured, examined, tested, and packed in accordance with contract documents, applicable list of drawings as listed in table I, and this specification.

Table I

LD number	Item nomenclature
LD 615246	Signal, Illumination, Marine, Mk 1 Mod 1 (Star)
LD 615260	Signal, Illumination, Marine, Mk 1 Mod 0 (Comet)
LD 615261	Signal, Illumination, Marine, Mk 1 Mod 2 (Chameleon)
LD 615262	Signal, Illumination, Marine, Mk 1 Mod 3 (Shower)
LD 615263	Signal, Illumination, Marine, Mk 1 Mod 4 (Occulting Chameleon)
LD 615264	Signal, Smoke, Marine, Mk 2 Mod 0
LD 615265	Signal, Smoke, Marine, Mk 2 Mod 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:43Z
Check the source to verify that this is the current version before use.

3.1.1 Preproduction sample. An acceptable preproduction sample of signals shall be furnished by the contractor. The signal, its components and all materials shall be of a single type, grade, class, size, and composition, and shall comply with all applicable provisions of this document. The sample shall be manufactured at the same location, using the same methods proposed for full production. The preproduction sample shall not be applied as part of the quantity specified for delivery by the contract or order.

3.2 Examinations.

3.2.1 Components, assemblies, and materials. All components, assemblies, and materials of the signal shall meet all requirements of their respective specifications or drawings.

3.2.1.1 Primer. The primer shall meet the requirements of the applicable drawing.

3.2.2 Signal. The signal shall meet the requirements of this specification.

3.3 Performance and product characteristics.

3.3.1 Performance.

3.3.1.1 Function. The signal shall meet the following requirements:

(a) Function: produce the specified colored display a minimum height of 500 feet above the surface.

(b) Propel the display to a minimum height of 300 feet above the surface

(c) Burning time shall be as specified in table II, as applicable.

3.3.2 Durability and environmental.

3.3.2.1 Forty-foot drop. The signal shall withstand forty-foot drop without burning or exploding.

3.3.2.2 Transportation vibration. The signal shall withstand transportation vibration without burning or exploding.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:43Z
Check the source to verify that this is the current version before use.

Table II

Item nomenclature	Burning time	
	Minimum (sec)	Maximum (sec)
Signal, Illumination, Marine, Mk 1 Mod 0 (Comet)	7	11
Signal, Illumination, Marine, Mk 1 Mod 1 (Star)	20	30
Signal, Illumination, Marine, Mk 1 Mod 2 (Chameleon)	7 (each star) 21 (total for signal)	-
Signal, Illumination, Marine, Mk 1 Mod 3 (Shower)	4	6
Signal, Illumination, Marine, Mk 1 Mod 4 (Occulting Chameleon)	7 (each occulting increment)	-
Signal, Smoke, Marine, Mk 2 Mod 0	20	30
Signal, Smoke, Marine, Mk 2 Mod 1	20	30

3.4 Workmanship. The product in all stages of production, shall be processed under such inspection control as to assure uniformity in quality. All components shall be free of chips, dirt, grit, or other foreign material. The cleaning methods used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents employed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:43Z
Check the source to verify that this is the current version before use.

4.1.1 Preproduction sample. The contractor shall deliver a sample of 80 signals of one Mk and Mod, to the destination specified (see 6.2), for examination, testing, evaluation, and acceptance as specified in 4.2 and table III as applicable. Any production of the item prior to production of an acceptable preproduction sample by the contractor or authorization by the procuring agency shall be at the risk of the contractor. The sample shall be prepared for delivery as specified in section 5.

Table III

INSPECTION PLANS

Plan[1]				Section reference	Tests	Preproduction sampling plans		Production lot sampling plans	
Pre-prod	A	B	C		1 2 3	Sample size	Acceptance criteria[2]	Sample size	Acceptance criteria[2]
X	X			4.3.2.1 3.3.2.1	Forty-foot drop test	5 Signals	Ac 0 Re 1	5 Signals	Ac 0 Re 1
X	X			4.3.2.2 3.3.2.2	Transportation vibration test	20 Signals	Ac 0 Re 1	MIL-STD-105 inspection level S-3	Ac 0 Re 1
				4.3.1.1 3.3.1.1	Function test		(a) Ac 1 Re 2 (b) Ac 0 Re 1 (c) Ac 3 Re 4		(a) AQL 2.5 (b) Ac 0 Re 1 (c) AQL 6.5
X	X	X		4.3.1.1 3.3.1.1	Function test	50 Signals	(a) Ac 3 Re 4 (b) Ac 0 Re 1 (c) Ac 7 Re 8	MIL-STD-105 inspection level II	(a) AQL 2.5 (b) Ac 0 Re 1 (c) AQL 6.5
X				4.3.2.3 3.3.1.1	Grouped data acceptance	70 Signals	(a) Ac 3 Re 4 (c) Ac 8 Re 9	None	None

[1] X - Denotes tests applicable to plan.
[2] Ac - Acceptance number; Re - Rejection number.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:43Z
Check the source to verify that this is the current version before use.

4.1.2 Production lot. A production lot shall consist of 501 to 10,000 signals of one Mk and Mod offered for delivery at one time including test samples. The samples shall not be applied as part of the quantity specified for delivery by the contract or purchase order. Sampling, examination, testing, and acceptance of production lots shall be performed as specified in the steps given below. Upon rejection of any lot, the next lot, in addition to the applicable plan, shall be examined, tested, and accepted for the requirement(s) which caused the rejection, unless the requirement(s) is included in the applicable inspection plan. Each production lot shall contain only primers produced during 1 year by one manufacturer.

STEP 1. Upon starting production or delivery of an acceptable preproduction sample, as applicable, the first two production lots shall be examined as required in 4.2, tested in sequence, and accepted in conformance with table III, plan A.

STEP 2. Production lots other than those defined in step 1 shall be examined as required in 4.2, tested in sequence, and accepted in conformance with table III, plan B, except that one lot shall be randomly selected from each five consecutive production lots and examined as required in 4.2, tested in sequence, and accepted in conformance with table III, plan A.

4.1.3 Test equipment and gages. All test equipment and gages shall be supplied and maintained by the contractor as specified in MIL-I-45607. The contractor shall maintain a calibration system as specified in MIL-C-45662.

4.1.3.1 Test conditions. The maximum allowable tolerances on test conditions, unless otherwise specified, shall be as follows:

(a) Temperature: ±3° centigrade, 5.4° Fahrenheit
(b) Relative humidity: +5 percent, -0 percent
(c) Vibration amplitude: ±10 percent
(d) Vibration frequency: ±2 percent.

4.1.4 Test and examination classification. The tests and examinations herein are classified in regard to relative importance as critical, major, or minor. Tests and examinations classified as critical are identified by the symbol (C) and major by the symbol (M). The number following the symbol indicates the serial number of the test or examination. Tests which are not annotated are classified minor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:43Z
Check the source to verify that this is the current version before use.

4.2 Examinations. Random samples shall be selected and examined as specified in 4.1.1 or 4.1.2 as applicable.

4.2.1 Components, assemblies, and materials examinations. Prior to final assembly, all components, assemblies, and materials used in the signal shall meet the requirements of 3.2.1.

4.2.1.1 Primer (M101). In the absence of previous inspection of the primer, inspection shall be performed as specified on the applicable drawing.

4.2.2 Signal examinations (M102). Examinations shall be made to ascertain that signals meet the requirements as specified on the applicable drawings. Only signals containing no defects shall be used for further testing or returned to the lot.

4.2.3 Packaging, packing, and marking examination (M103). The packaging, packing of the signals, and marking of the package and pack shall conform to section 5 of this specification.

4.3 Tests. Random samples of signals shall be selected and tested as specified in 4.1.1 or 4.1.2, as applicable.

4.3.1 Performance test.

4.3.1.1 Function test (M104). The signals shall be tested by firing vertically from a Rocket Pistol Mk 1 Mod 0 or Pyrotechnic Pistol AN-M8. Record the functioning, color, height at ejection of the display, and burning time. The height measurement is obtained using a computerized least squares fit or the following equivalent procedure:

The height of the display shall be measured by using two or more sighting devices, capable of measuring the vertical and horizontal angles within ±15 minutes from ground reference points to the ejected display height. The distance between the sighting devices and between the sighting devices and firing location shall be 400 to 1000 feet. The sighting devices shall be on the same horizontal plane with the launcher. (When the test location does not allow this, mathematically sound corrections can be made to the equation to compensate for ground elevation differences.) The following equation shall be used to calculate the display height.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:43Z
Check the source to verify that this is the current version before use.

$$H = \frac{\frac{D \sin A_1 \tan E_2}{\sin (A_1 + A_2)} + \frac{D(\sin A_2)(\tan E_1)}{\sin(A_1 + A_2)}}{2}$$

where:

H = Height
D = Distance between sighting devices
A_1 = Azimuth angle measured by the first device. The horizontal angle between the second device and the point of maximum height
A_2 = Azimuth angle measured by the second device. The horizontal angle between the first device and the point of maximum height
E_1 = Elevation angle measured by the first device. The vertical angle from the horizontal to the point of maximum height
E_2 = Elevation angle measured by the second device. The vertical angle from the horizontal to the point of maximum height.

Defectives are signals failing to meet the requirements of 3.3.1.1.

4.3.2 Durability and environmental tests.

4.3.2.1 Forty-foot drop test (C1). The signal shall be subjected to the forty-foot drop test prescribed by Test 103 of MIL-STD-331, except the signal shall be dropped free-fall without guidance system or associated equipment. Defectives are signals failing to meet the requirements of 3.3.2.1.

4.3.2.2 Transportation vibration test (C2). The signal shall be subjected to the vibration test prescribed by Test 104, procedure II of MIL-STD-331. Defectives are signals failing to meet the requirements of 3.3.2.2.

4.3.2.3 Grouped data acceptance. The data obtained from requirements (a) and (c) of the function test shall be grouped for the 70 signals composed of the following groups:

Group (a)	Twenty signals from the transportation vibration test of 4.3.2.2
Group (b)	Fifty signals from the function test of 4.3.1.1.

Acceptance shall be as specified in table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:43Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. In addition to the following, level A packaging shall be in accordance with the Code of Federal Regulations 49 CFR 171-178.

5.1.1.1 One signal shall be packaged in accordance with LD 615259.

5.2 Packing.

5.2.1 Level A. In addition to the following, level A packing shall be in accordance with the Code of Federal Regulations 49 CFR 171-178.

5.2.1.1 Twenty-four signals packaged as specified in 5.1.1.1 shall be packed in accordance with LD 615236.

5.3 Marking.

5.3.1 Special marking. In addition to the requirements of the contract or order the outer shipping container shall be marked in accordance with the Code of Federal Regulations 49 CFR 171-178 for class B explosives (Special Fireworks).

5.3.2 Normal marking of the package and pack. In addition to any requirements of the contract or order, packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The signals are intended to produce a visual smoke or flame of the required color for signaling purposes.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification

(b) Mark, mod, and color required

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:43Z
Check the source to verify that this is the current version before use.

(c) Quantity of signals being ordered

(d) Production lots size

(e) QE laboratory coordinating surveillance

(f) Test facility where preproduction sample is to be shipped for evaluation (4.1.1).

(g) That the safety precaution requirements of the Contractor's Safety Manual for Ammunition, Explosives and Related Dangerous Material, DOD 4145.26M, are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of Ammunition Ashore, OP 5, should be made applicable.

6.2.2 Contract data requirements. For the information of contractors and contracting officers, any of the data specified in (a) subparagraph below, (b) applicable documents listed in section 2 of this specification, or (c) referenced lower-tier documents, need not be prepared for the Government and shall not be furnished to the Government unless justified and specified in the contract or order. The data to be furnished shall be listed on DD Form 1423, which shall be attached to and made a part of the contract or order and described in a Data Item Description (DID) referenced by number in block 4 or 5 of DD Form 1423.

6.2.2.1 Data cards. If applicable (see 6.2.2), the contractor shall fill out the following data card:

Form No.	Title	DID No.
DD Form 1650	Ammunition Data Card	DI-E-2001

The contractor shall fill in all the information required by the data card for each of the following components:

(a) Primer

(b) Candle

(c) Parachute.

The Government inspector shall endorse the cards and distribute one each as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:43Z
Check the source to verify that this is the current version before use.

(a) Procuring activity

(b) SPCC (Code 730), Mechanicsburg, Pa. (if different than the procuring activity)

(c) The QE laboratory coordinating surveillance (see 6.2)

(d) The designated test activity (see 6.2).

6.3 Conflicting requirements. Conflicting requirements arising between this specification and any specification or publication listed herein shall be referred in writing to the procuring agency for clarification.

Custodian:
NAVY - OS

Preparing activity:
NAVY - OS
(Project No. 1370-N401)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:43Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-S-18568B(OS), Signal, Illumination, Marine, Mk 1 Mods 0, 1, 2, 3, and 4 and Signal, Smoke, Marine, Mk 2 Mods 0 and 1

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:43Z
Check the source to verify that this is the current version before use.